	GLOBAL ECONOMICS **Grade: 11th - Learning Evidence 3** **Statistical models and research planning** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Apply continuous probability models and sampling distributions to investigate economic and financial questions, justify statistical methods, and communicate evidence-based conclusions.	**Assessment criteria:** C8: Uses Student's t-distribution tables, degrees of freedom, and one-tailed or two-tailed areas to determine critical values, central probabilities, and tail probabilities or probability ranges. C9: Calculates and compares critical values and probability areas from the standard normal and Student's t-distributions, selecting and justifying the appropriate distribution according to whether the population standard deviation is known or estimated from the sample. C10: Integrates continuous probability modelling, simulation, sampling analysis, and portfolio-risk measures to investigate an economic or financial question, justifying the selected methods and communicating results, assumptions, limitations, and evidence-based conclusions.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

Partial credit may be awarded when the correct procedure is clearly demonstrated, even if arithmetic errors are made. No points will be awarded for answers without a demonstrated procedure. A minimum of **4 points** is required to pass each criterion. A 1-point penalty will be applied to each criterion if the student's name is not written on the assessment or if the assessment is submitted late.

№	C8
1–5	5
6–10	5
Total	**10**

№	C9
11–13	6
14–16	6
Total	**12**

Conventions

Models. C8 (1–10): $X = m + aT$. C9 (11–16): $X_T = m + aT$, $X_N = m + aZ$, where $T \sim t_{n-1}$, $Z \sim N(0,1)$, n is sample size, m is location, and $a > 0$ is scale.

Standardization. Subtract m and divide by a:

$$T = \frac{X_T - m}{a}, \qquad Z = \frac{X_N - m}{a}.$$

For inverse problems, transform back using $x = m + at$ or $x = m + az$.

C9 comparisons. For every problem, calculate r_T, r_N, and

$$E_A = |r_T - r_N|, \qquad E_{Pt} = \frac{100E_A}{|r_T|}, \qquad E_{Pn} = \frac{100E_A}{|r_N|}.$$

1. Courier arrival deviations [C8-1]

C8 – Direct left tail. A sample of $n = 10$ courier trips has arrival-time deviation from schedule, in minutes, with mean $m = 1$ and approximate standard deviation $a = 3$. Use $X = 1 + 3T$, $T \sim t_{n-1}$. Find $P(X < -4.499)$ and interpret the threshold. **[1]**

2. Freight cost deviations [C8-1]

C8 – Direct right tail. A sample of $n = 13$ consignments has cost deviation from budget, in dollars, with mean $m = -4$ and approximate standard deviation $a = 8$. Use $X = -4 + 8T$, $T \sim t_{n-1}$. Find $P(X > -10.984)$ using a complement, and explain why it exceeds 0.5. **[1]**

3. Warehouse energy deviations [C8-1]

C8 – Direct interval above the mean. Over $n = 17$ days, electricity-use deviation from forecast, in kWh, has mean $m = 2$ and approximate standard deviation $a = 6$. Use $X = 2 + 6T$, $T \sim t_{n-1}$. Find $P(7.19 < X < 12.476)$. **[1]**

4. Crop shipment deviations [C8-1]

C8 – Direct interval below the mean. A sample of $n = 8$ crop shipments has mass deviation from forecast, in tonnes, with mean $m = -1$ and approximate standard deviation $a = 3$. Use $X = -1 + 3T$, $T \sim t_{n-1}$. Find $P(-8.095 < X < -3.688)$. **[1]**

5. Retail revenue alerts [C8-1]

C8 – Direct unequal combined tails. Over $n = 7$ weeks, revenue deviation from plan, in thousands of dollars, has mean $m = 2$ and approximate standard deviation $a = 4$. Use $X = 2 + 4T$, $T \sim t_{n-1}$. Find the probability of an alert: $X < -5.772$ or $X > 7.76$. **[1]**

6. Operating cost cutoff [C8-1]

C8 – Inverse lower-tail cutoff. Over $n = 15$ production days, operating-cost deviation from budget, in dollars, has mean $m = -3$ and approximate standard deviation $a = 5$. Use $X = -3 + 5T$, $T \sim t_{n-1}$. Find x such that $P(X < x) = 0.10$. **[1]**

7. Loading time cutoff [C8-1]

C8 – Inverse upper-tail cutoff. A sample of $n = 21$ loading jobs has duration deviation from target, in minutes, with mean $m = 2$ and approximate standard deviation $a = 7$. Use $X = 2 + 7T$, $T \sim t_{n-1}$. Find x such that $P(X > x) = 0.025$. **[1]**

8. Demand deviation interval [C8-1]

C8 – Inverse central interval. Over $n = 12$ weeks, demand deviation from forecast, in tonnes, has mean $m = 4$ and approximate standard deviation $a = 9$. Use $X = 4 + 9T$, $T \sim t_{n-1}$. Find $L < U$ enclosing the central 90% of deviations with equal excluded tails. **[1]**

9. Package mass alerts [C8-1]

C8 – Inverse equal-tail alert limits. A sample of $n = 9$ packages has mass deviation from target, in grams, with mean $m = -0.4$ and approximate standard deviation $a = 1.2$. Use $X = -0.4 + 1.2T$, $T \sim t_{n-1}$. Find two alert limits leaving a combined outside probability of 0.05, split equally. **[1]**

10. Dispatch time alerts [C8-1]

C8 – Inverse unequal-tail alert limits. A sample of $n = 16$ dispatches has departure-time deviation from schedule, in minutes, with mean $m = -2$ and approximate standard deviation $a = 6$. Use $X = -2 + 6T$, $T \sim t_{n-1}$. Find L, U with $P(X < L) = 0.01$ and $P(X > U) = 0.10$. **[1]**

11. Sales revenue shortfalls [C9-2]

C9 – Direct left tail. Over $n = 10$ weeks, sales-revenue deviation from target, in thousands of dollars, has mean $m = -2$ and approximate standard deviation $a = 5$. Compare $X_T = -2 + 5T$, $T \sim t_{n-1}$, with $X_N = -2 + 5Z$. Find $P(X < -11.165)$. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}, and interpret the comparison. **[2]**

12. Delivery schedule deviations [C9-2]

C9 – Direct right tail. A sample of $n = 15$ deliveries has arrival-time deviation from schedule, in minutes, with mean $m = 3$ and approximate standard deviation $a = 4$. Compare $X_T = 3 + 4T$, $T \sim t_{n-1}$, with $X_N = 3 + 4Z$. Find $P(X > -0.472)$. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}, and interpret the comparison. **[2]**

13. Fuel forecast deviations [C9-2]

C9 – Direct interval crossing the mean. Over $n = 13$ operating days, fuel-use deviation from forecast, in litres, has mean $m = 1$ and approximate standard deviation $a = 6$. Compare $X_T = 1 + 6T$, $T \sim t_{n-1}$, with $X_N = 1 + 6Z$. Find $P(-7.136 < X < 14.074)$. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}, and interpret the comparison. **[2]**

14. Production cost savings [C9-2]

C9 – Inverse lower-tail cutoff. A sample of $n = 8$ batches has cost deviation from budget, in dollars, with mean $m = -1$ and approximate standard deviation $a = 2.5$. Compare $X_T = -1 + 2.5T$, $T \sim t_{n-1}$, with $X_N = -1 + 2.5Z$. Find the cutoff x below which 5% of deviations fall. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}, and interpret the comparison. **[2]**

15. Order processing cutoff [C9-2]

C9 – Inverse upper-tail cutoff. A sample of $n = 17$ orders has processing-time deviation from target, in minutes, with mean $m = 2$ and approximate standard deviation $a = 3.5$. Compare $X_T = 2 + 3.5T$, $T \sim t_{n-1}$, with $X_N = 2 + 3.5Z$. Find the cutoff x exceeded by 10% of deviations. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}, and interpret the comparison. **[2]**

16. Cargo mass deviation interval [C9-2]

C9 – Inverse central interval. A sample of $n = 10$ cargo loads has mass deviation from target, in kg, with mean $m = 1.5$ and approximate standard deviation $a = 4$. Compare $X_T = 1.5 + 4T$, $T \sim t_{n-1}$, with $X_N = 1.5 + 4Z$. Find the central 95% interval with equal excluded tails under each model. Calculate E_A, E_{Pt}, and E_{Pn} separately for each endpoint, and interpret. **[2]**

17. Research option [C10-6]

Exam: Select one model and describe plausible real-life data in 1–2 sentences. **[6]**

- **(a)** Linear probability density over a finite interval.
- **(b)** Exponential distribution (waiting times).
- **(c)** Normal distribution (measurements).

Online report: Submit a PDF through **Open** using real economic or financial data and the selected model. Include simulation, sampling, portfolio risk, methods, assumptions, limitations, and evidence-based conclusions. The report provides additional C10 evidence.